

Course Introduction

* we will cover over 40 AWS services (out of 200+ in AWS)

* AWS/IT Beginners welcome!

* learn by doing - key learning technique!

This course mixes both theory and hands on

Sample questions :-

Which AWS service would simplify the migration of a database to AWS?

- A) AWS Storage Gateway <= we will learn
- B) AWS Database Migration Service <= correct answer
- C) Amazon EC2 <= we will learn
- D) Amazon AppStream 2.0 <= distractor (over 200 services in AWS)

Your AWS Certification journey

Foundational

Learn AWS Cloud fundamentals and core concepts.
No prior experience needed.



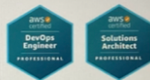
Associate

Validate core AWS skills. Prior cloud or IT experience recommended.



Professional

Master advanced AWS architecture and solutions.
2+ years AWS experience required.



Specialty

Demonstrate expertise in specific AWS technologies and services.



Estimated Cost for this Course

- AWS will give you \$100 of free credits when you create your account (can go up to \$200)

- Creating resources in the cloud will cost money but they will use the credits you'll have (if not I'll let you know)

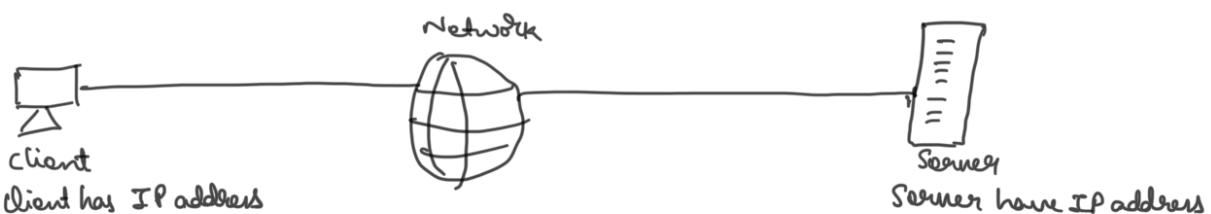
- You can read more about the AWS Free Tier at: <https://aws.amazon.com/free/>

	\$12.70
Registrar	\$12.00
Route53	\$0.50
Config	\$0.18
KMS	\$0.02
Other Services	\$0.00
Total	\$12.70

Creating AWS account in personal computer.

Section 3:- what is cloud computing?

How websites work :-



* web browser want to access to server to visualize a website. what we are going to do as

Client is use a network. A Network between ourselves and a Server. Client will find the network. use Network to route the packets, then data into the Server. Then the Server will send to us. we get the response and we can view the website.

* The client to find the server and server to find client, you need to have IP addresses.

The idea is when you use a IP address you can send a request, whenever you want, to server you want and server can know how to find you back.

Ex:- This is similar to writing a letter to your friend. Writing a letter that would be data, and you would be a client, then when you send a letter you put in your mailbox. And network will be network to post office, then post office will use network and address you put on letter to route your letter to destination which is server. And correspondent wants to reply you back, they can use address to envelope and if your correspondent wants to reply you back. They can use address you put on back of envelope to write you back and again, and again use same network to get letter back to you.

What is a server composed of?

* A server is going to contain a CPU, CPU is a little piece that will be doing some computation.

* A server also needs RAM or memory. This is going to be very very fast memory which will allow us to store information and retrieve it very quickly.

CPU + Memory = Brain (when you are thinking, you are actually making computation)

* Now, we need to have long term storage of data. Obviously it is still in our brain as humans, but in computers we have included special storage files. If we wanted to store data in more structured way then database.

* Finally in server we are going to have some networking aspects. (Routers, Switch, DNS Server).

So in server, we are aspect of compute, memory, storage, maybe your server something in database. We have networking aspect.

What is a server composed of?

- Compute: CPU
- Memory: RAM



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- Storage: Data



cloud is giving us on demand.

- Database: Store data in a structured way



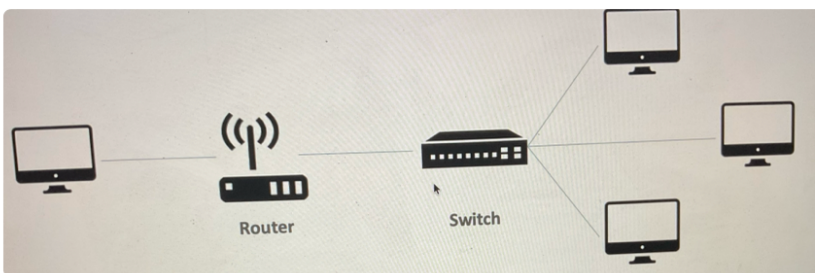
- Network: Routers, switch, DNS server



Network:- Is bunch of cables, routers and servers connected with each other.

Router:- A networking device that forwards data packets between computer networks. They know where to send your packets on Internet!

Switch:- Takes a packet and send it to correct server / client on your network.



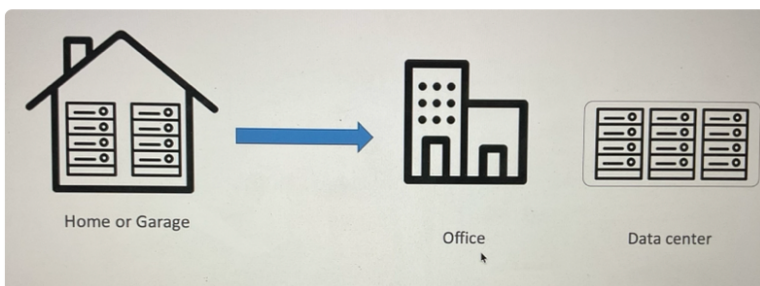
Client will send a data to Router, the Router will find its way all the way to Switch, Switch will know which computer in your network to send the data.

Traditionally In older days:-

They used to do in home or garage, so they would literally go to store, buy server and put the server in home. How google was made. Google was started in a garage.

As your website grow you need to add more servers to serve that demand, your home start to be filled with servers. Your company is growing so you need to build your own office and allocate a special room (which is going to be called data center).

In data center you will be purchasing more and more servers. But they are few problems.



- Problems
- 1) pay for rent for data center
 - 2) pay for power supply, cooling, maintenance
(It required because servers heat a lot sometimes they break down the maintenance and if you want to add or replace servers).
 - 3) Adding and replacing hardware takes time

4) Scaling is limited. / 5) Hire 24/7 team to monitor the infrastructure.

6) How to deal with disasters? (earthquakes, power shutdown, fire) / 7) Can we externalize all this?